

Chapter 1

Stage A Molnar Single-Notch Benchmark

1.1 Objective and Decision Boundary

Stage A establishes whether the Molnar and Gravouil single-notch phase-field benchmark is a defensible baseline for later Abaqus adaptive-remeshing work. The aim is deliberately narrower than the full thesis workflow: reproduce an established staggered phase-field implementation, quantify the reconstructed single-notch response, and decide whether the evidence is strong enough to authorize Stage B uniform-reference studies.

The current Stage A status is:

```
candidate v2: technically passed
scientific review: incomplete
SDV15: completed-increment irreversibility violation
Gate A3: reference_data_insufficient
new solution run: not authorized
```

This chapter therefore treats candidate v2 as a technically reproducible and useful reproduction baseline, but not as a Gate A3 pass. Supervisor approval is required before any Stage B, uniform-reference, MISESeri, remeshing, or state transfer work begins.

1.2 Benchmark Choice

The Molnar single-notch benchmark was selected because it connects directly to the thesis requirement for an Abaqus UEL/UMAT phase-field workflow. The source package contains a staggered displacement/phase implementation, a visualization layer through state variables, and single-notch examples suitable for checking reaction force, crack-path geometry, and phase-field/history behavior before introducing adaptive remeshing.

The benchmark is also a useful stress test for thesis discipline. It separates technical execution from scientific validation: a model can compile, solve, and produce a plausible crack while still failing a strict scientific gate because the RF-U reference, tolerances, or irreversibility interpretation remain unresolved.

1.3 Formulation Convention

The phase-field convention used throughout Stage A is:

- $d = 0$: intact material.
- $d = 1$: fully broken material.

Candidate v2 preserves the published Molnar source formulation. The generated source copy `models/generated/molnar_gravouil_2017/paper_matched_single_notch_v2/SingleNotch_v2.for` is copied from the preserved source with only the physical-element count `N_ELEM=33852` updated for the reconstructed mesh. The original source is preserved at `models/baseline_original/molnar_gravouil_2017/02_Single_Notch_Tension/SingleNotch.for`.

1.4 Staggered Phase/Displacement Method

The implementation follows a staggered solution structure with displacement and phase-field layers. The displacement field is solved in the mechanical layer, and the phase-field layer updates the scalar damage variable using the stored history information. Visualization variables are copied into a companion UMAT/CPS4 layer; those visualization fields are evidence outputs, not separate governing fields.

For thesis use, three distinctions must remain visible:

- The published formulation is the Molnar and Gravouil staggered phase-field UEL/UMAT implementation.
- The reconstructed implementation is the project candidate-v2 paper-matched deck generated from documented geometry, loading, material, and mesh assumptions.
- The approximate Fig. 7 reference is a local digitization of the published curve, not exact author data and not a supervisor-approved final tolerance basis.

1.5 Geometry, Material, and Loading

Candidate v2 reconstructs a 1.0 mm $\times$ 1.0 mm single-edge-notched plate with a left-edge split-node notch of length 0.5 mm at $y = 0$. The model uses plane strain and unit thickness. Material parameters are $E = 210$ kN/mm², $\nu = 0.3$, $G_c = 0.0027$ kN/mm, and $l_c = 0.0075$ mm.

The load angle is 90°. The loading is applied in two direct-increment steps:

Step	Start U_2 (mm)	End U_2 (mm)	Increment (mm)
Step 1	0.0000	0.0050	1.0×10^{-5}
Step 2	0.0050	0.0067	1.0×10^{-5}

1.6 Paper-Matched Reconstruction

The reconstruction is documented in the project configuration and derived reference records. Candidate v2 restores the intended loading arithmetic, implements a split-node notch, preserves the layered UEL/UMAT architecture, and uses the source property layout from the preserved Molnar files. Static validation passed, and the mesh-quality preflight classified the model as `mesh_quality_preflight_pass`.

The mesh has 33852 physical elements and 101556 layered elements. The local mesh size is $h = 0.001$ mm, giving $h/l = 0.1333333333$. The maximum element size is 0.025 mm, the maximum neighboring-size ratio is 1.5, and the maximum aspect ratio is 25.000000000001368. The high-aspect-ratio elements are documented as reconstruction limitations outside the fracture-process corridor.

1.7 Technical Validation

The single authorized serial HPC baseline job was 1374864.mm`master02`, submitted from revision 711dd495bdc830d695f9d7e5. It completed with `PBS Exit_status=0`; Abaqus returned code zero; the `.sta`, `.msg`, `.dat`, and scratch `.odb` outputs existed; and the `.sta` reported successful analysis completion. The technical classification is `paper_matched_v2_technical_pass`.

The run used one CPU, 32 GB requested memory, and a 24-hour walltime request. Recorded resource use was walltime 00:38:38, CPU time 00:35:52, CPU percent 95, memory 2970760 kb, virtual memory 3565868 kb, and an Abaqus-reported peak memory of about 2 GB.

1.8 Numerical Results

Table 1.1: Stage A candidate-v2 numerical evidence. Values are drawn from committed extraction, scientific-review, and targeted-diagnostic evidence.

Metric	Value	Evidence
Peak RF2	0.761702 kN	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Peak displacement	$U_2 = 0.006110$ mm	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Peak-force error	0.064519 relative	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Peak-displacement error	0.041257 relative	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Pre-peak RMSE	0.044136 kN	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Post-peak RMSE	0.348093 kN	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Full-overlap NRMSE	0.245705	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Tangent-stiffness error	about +51.4%	runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md
Common-domain area error	+0.193324 relative	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Crack extension, $SDV15 \geq 0.80$	0.0555 mm beyond notch tip; 0.0570 mm total connected path	runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md

Metric	Value	Evidence
Crack extension, $SDV15 \geq 0.90$	0.0525 mm beyond notch tip; 0.0530 mm total connected path	runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md
Crack extension, $SDV15 \geq 0.95$	0.0505 mm beyond notch tip; 0.0510 mm total connected path	runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md
Crack extension, $SDV15 \geq 0.99$	0.0465 mm beyond notch tip; 0.0460 mm total connected path	runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md
SDV15 maximum overshoot	1.005600	runs/hpc/paper_matched_single_notch_v2/scientific_review/sdv_irreversibility_metrics.json
Accepted-increment decreasing transitions	2184	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
Transitions above 10^{-6}	2131	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmmaster02/postprocessing_completed_increment_replay_time_aligned/severity_audit/sdv15_completed_increment_severity_summary.json
Transitions above 10^{-5}	1362	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmmaster02/postprocessing_completed_increment_replay_time_aligned/severity_audit/sdv15_completed_increment_severity_summary.json
Maximum decrease	$2.2384088238425193 \times 10^{-4}$	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
Mean decrease	$2.8434597987446906 \times 10^{-5}$	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md

Metric	Value	Evidence
Median decrease	$1.3940791172561973 \times 10^{-5}$	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
Unique affected element/IP locations	138	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
SDV16 decreases	0	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
Diagnostic RF–U perturbation	maximum normalized RF2 difference $6.54442468760855 \times 10^{-13}$; maximum U2 difference $4.992967844730245 \times 10^{-15}$ mm	runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md
Solver walltime and memory	baseline PBS wall-time 00:38:38, memory 2970760 kb, Abaqus peak memory about 2 GB	runs/hpc/paper_matched_single_notch_v2/scientific_review/solver_resource_metrics.json

1.9 RF–U Interpretation

The RF–U comparison is mixed. The peak force scale is close enough to justify continued review, but the full curve is not within a strict provisional working gate. The pre-reference-peak RMSE is 0.044136 kN, while the post-reference-peak RMSE is 0.348093 kN. This makes the post-peak mismatch the dominant scientific limitation. The comparison uses the approximate digitized Fig. 7 red dashed $l_c = 0.0075$ mm reference, which has estimated digitization uncertainty of about ± 0.00004 mm and ± 0.015 kN.

1.10 Crack-Path Interpretation

The final high-damage band is connected and approximately horizontal at thresholds from $\text{SDV15} \geq 0.80$ through $\text{SDV15} \geq 0.99$. At the provisional $\text{SDV15} \geq 0.95$ threshold, the final matched contour contains 193 element-mean damaged elements and a connected high-damage extension of about 0.0505 mm beyond the notch tip. This supports qualitative crack-direction agreement, but it remains threshold-dependent and does not close Gate A3 by itself.

1.11 SDV15, SDV16, and Irreversibility

In the project convention, SDV15 is the phase-field visualization variable and SDV16 is the history-like variable used to monitor monotonic driving behavior. The original ODB scan found SDV16 monotone and SDV15 nonmonotone in retained visualization frames. A later targeted diagnostic r2 run was non-intrusive by RF–U comparison and enabled a no-solution completed-increment replay of the existing trace rows.

The completed-increment replay kept the last U1 stage-101 call for each (KSTEP, KINC, physical element, source – storage). It processed 627304 trace rows, retained 209152 U1 stage-101 rows, formed 101840 final increment states over 152 element/IP sequences, and found 2184 unique completed-increment SDV15 decreasing transitions.

The severity audit classifies this as `sdv15_completed_increment_irreversibility_violation` under the provisional 10^{-6} materiality tolerance. SDV16 has zero decreases over the same final-increment sequences.

1.12 Scientific Limitations

The current evidence supports the following restrained interpretation:

- Candidate v2 is technically reproducible.
- Crack propagation direction is qualitatively correct.
- RF–U agreement is mixed, and post-peak disagreement is substantial.
- SDV16 remains monotone in the checked sequences.
- SDV15 decreases between accepted increments.
- The targeted diagnostic instrumentation is non-intrusive.
- Strict pointwise phase-field irreversibility is not satisfied.
- Candidate v2 remains a useful reproduction baseline for supervisor review.
- Supervisor approval is required before Stage B or any downstream adaptive-remeshing work.

1.13 Gate A3 Boundary

Gate A3 remains `reference_data_insufficient`. Candidate v2 has not passed Gate A3. No new solution run, candidate v3, irreversibility-enforced formulation, uniform-reference study, MISESeri workflow, adaptive-remeshing run, or state-transfer task is authorized by the present evidence.